Experiment 9

Docker is an open-source containerization platform that allows developers to build, deploy, and run applications in isolated environments known as containers. These containers package the application along with all its dependencies, ensuring consistency and portability across different systems.

1. Installation of Docker

Docker is installed on Windows using Docker Desktop, which provides a complete environment to build and run containers. It uses **Windows Subsystem for Linux (WSL 2)** as its backend to enable Linux-based container execution.

The installation process includes:

- Downloading Docker Desktop from the official source
- Installing and enabling WSL 2
- Configuring Docker to use the WSL 2 backend
- Starting Docker Desktop service

Once installed, Docker runs in the background and provides a command-line interface for executing Docker commands.

2. Docker Architecture

Docker follows a client-server architecture consisting of:

- **Docker Client**: Used to interact with Docker through commands
- **Docker Daemon**: Runs in the background and manages containers
- **Docker Objects**: Includes images, containers, volumes, and networks

3. Verification of Docker Installation

The command:

```
docker --version
```

is used to verify whether Docker has been installed successfully.

Explanation:

- Displays the installed Docker version and build number
- Confirms that the Docker CLI is properly configured
- Ensures the system is ready to execute Docker commands

4. Docker Information Command

The command:

docker info

provides detailed information about the Docker environment, including both clientside and server-side configurations.

Client Information

- Docker version
- Context (e.g., desktop-linux)
- Debug mode status
- Installed plugins such as buildx and compose

Server Information

- Number of containers (running, paused, stopped)
- Number of images available
- Server version
- Storage driver (e.g., overlay2) used for managing container data
- Logging driver (e.g., json-file) used for handling logs
- Cgroup driver for resource management
- Available runtimes (e.g., runc, nvidia)
- Networking options and plugins
- Docker root directory where all container data is stored

5. System and Environment Details

The docker info command also provides system-level details such as:

- Operating system and architecture
- CPU count and total memory
- Kernel version
- Docker Desktop configuration

These details help in understanding the environment in which Docker is running and are useful for troubleshooting and performance analysis.

Conclusion

Thus, installing Docker and executing commands like docker --version and docker info helps verify the setup and provides detailed insights into Docker's configuration and working. These commands are essential for ensuring that Docker is correctly installed and functioning properly before deploying containers.

```
Administrator: Command Pro

Microsoft Windows [Version 10.0.26100.7623]
(c) Microsoft Corporation. All rights reserved.

C:\Users\15L>docker --version
Docker version 29.2.1, build a5c7197

C:\Users\15L>
```

```
C:\Users\15L>docker info
Client:
 Version:    29.2.1
 Context:    desktop-linux
 Debug Mode: false
 Plugins:
  agent: create or run AI agents (Docker Inc.)
    Version:  v1.27.1
    Path:     C:\Program Files\Docker\cli-plugins\docker-agent.exe
  ai: Docker AI Agent - Ask Gordon (Docker Inc.)
    Version:  v1.18.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-ai.exe
  buildx: Docker Buildx (Docker Inc.)
    Version:  v0.31.1-desktop.1
    Path:     C:\Program Files\Docker\cli-plugins\docker-buildx.exe
  compose: Docker Compose (Docker Inc.)
    Version:  v5.0.2
    Path:     C:\Program Files\Docker\cli-plugins\docker-compose.exe
  debug: Get a shell into any image or container (Docker Inc.)
    Version:  0.0.47
    Path:     C:\Program Files\Docker\cli-plugins\docker-debug.exe
  desktop: Docker Desktop commands (Docker Inc.)
    Version:  v0.3.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-desktop.exe
  extension: Manages Docker extensions (Docker Inc.)
    Version:  v0.2.31
    Path:     C:\Program Files\Docker\cli-plugins\docker-extension.exe
  init: Creates Docker-related starter files for your project (Docker Inc.)
    Version:  v1.4.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-init.exe
  mcp: Docker MCP Plugin (Docker Inc.)
    Version:  v0.40.1
    Path:     C:\Program Files\Docker\cli-plugins\docker-mcp.exe
  model: Docker Model Runner (Docker Inc.)
    Version:  v1.1.1
    Path:     C:\Program Files\Docker\cli-plugins\docker-model.exe
  offload: Docker Offload (Docker Inc.)
    Version:  v0.5.56
    Path:     C:\Program Files\Docker\cli-plugins\docker-offload.exe
  pass: Docker Pass Secrets Manager Plugin (beta) (Docker Inc.)
    Version:  v0.0.24
    Path:     C:\Program Files\Docker\cli-plugins\docker-pass.exe
  sandbox: Docker Sandbox (Docker Inc.)
    Version:  v0.12.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-sandbox.exe
  sbom: View the packaged-based Software Bill Of Materials (SBOM) for an image (Anchore Inc.)
    Version:  0.6.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-sbom.exe
  scout: Docker Scout (Docker Inc.)
    Version:  v1.20.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-scout.exe
```

```
Server:
 Containers: 25
  Running: 0
  Paused: 0
  Stopped: 25
 Images: 9
 Server Version: 29.2.1
 Storage Driver: overlay2
  Backing Filesystem: extfs
  Supports d_type: true
  Using metacopy: false
  Native Overlay Diff: true
  userxattr: false
 Logging Driver: json-file
 Cgroup Driver: cgroupfs
 Cgroup Version: 2
 Plugins:
  Volume: local
  Network: bridge host ipvlan macvlan null overlay
  Log: awslogs fluentd gcplogs gelf journald json-file local splunk syslog
 CDI spec directories:
  /etc/cdi
  /var/run/cdi
 Discovered Devices:
  cdi: docker.com/gpu=webgpu
 Swarm: inactive
 Runtimes: io.containerd.runc.v2 nvidia runc
 Default Runtime: runc
 Init Binary: docker-init
 containerd version: dea7da592f5d1d2b7755e3a161be07f43fad8f75
 runc version: v1.3.4-0-gd6d73eb8
 init version: de40ad0
 Security Options:
  seccomp
   Profile: builtin
  cgroupns
 Kernel Version: 6.6.87.2-microsoft-standard-WSL2
 Operating System: Docker Desktop
 OSType: linux
 Architecture: x86_64
 CPUs: 20
 Total Memory: 15.5GiB
 Name: docker-desktop
 ID: 8426088b-7a55-4da0-a9e7-301b5465a3fd
 Docker Root Dir: /var/lib/docker
 Debug Mode: false
 HTTP Proxy: http.docker.internal:3128
 HTTPS Proxy: http.docker.internal:3128
 No Proxy: hubproxy.docker.internal
 Labels:
```